

TRACEABILITY OF SPECTROPHOTOMETRIC MEASUREMENTS TO THE HIGHEST STANDARD OF CALIBRATION

R.D. Pepenene, E.M. Coetzee

National Metrology Institute of South Africa (NIMSA)
Private Bag X34, Lynnwood Ridge, Pretoria, 0040, South Africa
rpepenene@nmisa.org
Phone: 012 841 4260 Fax: 012 841 3131

ABSTRACT

UV-Visible spectrophotometers are versatile instruments that measure a variety of different samples in diverse laboratory settings. The technique is mainly used quantitatively (although some qualitative analysis can also be performed). For any type of critical determination, whether it be clinical, pharmaceutical or industrial quality control, environmental analysis or research, it is essential that the instrument is verified to ensure that its measurements are traceable to appropriate measurement standards maintained at the National Metrology Institute. Traceability ensures equivalent measurements are made regardless of the instrument or the region of the world where the measurement is made.

In this paper, the principal calibration of secondary reference materials (Holmium Oxide filters and Neutral density filters) used in the verification of laboratory spectrophotometers for accurate absorption spectroscopy will be discussed. Other tests (e.g., the linearity of the absorption scale, baseline measurement and absorbance stability) will be used to test the performance of the Transfer Reference Spectrophotometer used to maintain and disseminate the national transmittance/absorbance scale.

Keywords: UV-Visible Spectrophotometer; Traceability of Measurement Results; Certified Reference Materials (Holmium Oxide Filter and Neutral Density Filter).

1. INTRODUCTION

Through the years, ultraviolet and visible spectrophotometry has been the method of choice in most laboratories concerned with the identification and measurement of organic and inorganic compounds in a wide range of products and processes e.g. in foodstuffs, pharmaceuticals, mineral oils and in paint. For any type of critical determination, it is essential that the wavelength accuracy and the photometric accuracy (absorbance accuracy) of an instrument are verified for the accurate peak wavelength determination and correct absorbance measurement [1]. Secondary reference materials used for this purpose need to be certified by either a national metrology institute (NMI) or an accredited calibration laboratory to ensure traceability of the measurement results to the highest standard. It is within the National Metrology Institute of South Africa (NMISA's) mission to maintain the national transmittance/absorbance scale, and disseminate it for the benefit of laboratories, industries, and others who need the highest accuracy measurements.

2. ABSORPTION SPECTROSCOPY

Ultraviolet-visible spectrophotometry (UV/Vis) refers to absorption spectroscopy in the ultraviolet-visible spectral region. This method of analysis is based upon the fact that the absorbance of the compounds is measured precisely at the specific wavelength of the absorption peak (λ_{max}). The Beer-Lambert law states that the amount of light radiation absorbed by a sample is directly related to the concentration of the coloured compound in the sample [2]. Thus, for a fixed path length, UV/V is spectroscopy can be used to determine the concentration of the absorber in a solution (**equation 1**). This makes it important that the wavelength accuracy and the photometric accuracy (absorbance accuracy) of an instrument be checked for the accurate peak wavelength determination and absorbance measurement.

$$A = \log_{10} \left(\frac{I_0}{I_1} \right) = \varepsilon \cdot c \cdot l \dots \dots \dots (1)$$

Where A is the measured absorbance, in Abs, I_0 is the intensity of the incident light at a given wavelength, I_1 is the transmitted intensity, l the path length through the sample, and c the concentration of the absorbing species. For each species and wavelength, ε is a constant known as the molar absorptivity which is a measurement of how strongly a chemical species absorbs light at a given wavelength).

3. UV-VISIBLE SPECTROPHOTOMETER

The Reference Transmittance Spectrophotometer (RTS) housed in the NMISA is the national measurement standard which establishes the national scale for regular spectral transmittance measurements at controlled conditions of humidity and temperature. This instrument is linked to the primary transmittance scale by means of optical filter reference materials manufactured and calibrated at the Nation Physical Laboratory (NPL) in the United Kingdom. It consists of a source of radiant energy, a double dispersing system to provide monochromatic radiation, and a detector system to measure the amount of radiation through the instrument [3] (see **figure 1**). The readout consists of adjusting the response to a value of 100% for the reference or standard. With the sample in place, the response of the detector represents the percentage of the sample's transmittance to that of the reference.

The path of radiation is split into two parts within the instrument to provide a sample beam and a reference beam. When a sample is placed in the sample beam, the equality of the two beams is broken and the detector senses the difference and relates that to the transmittance of the sample at that wavelength.

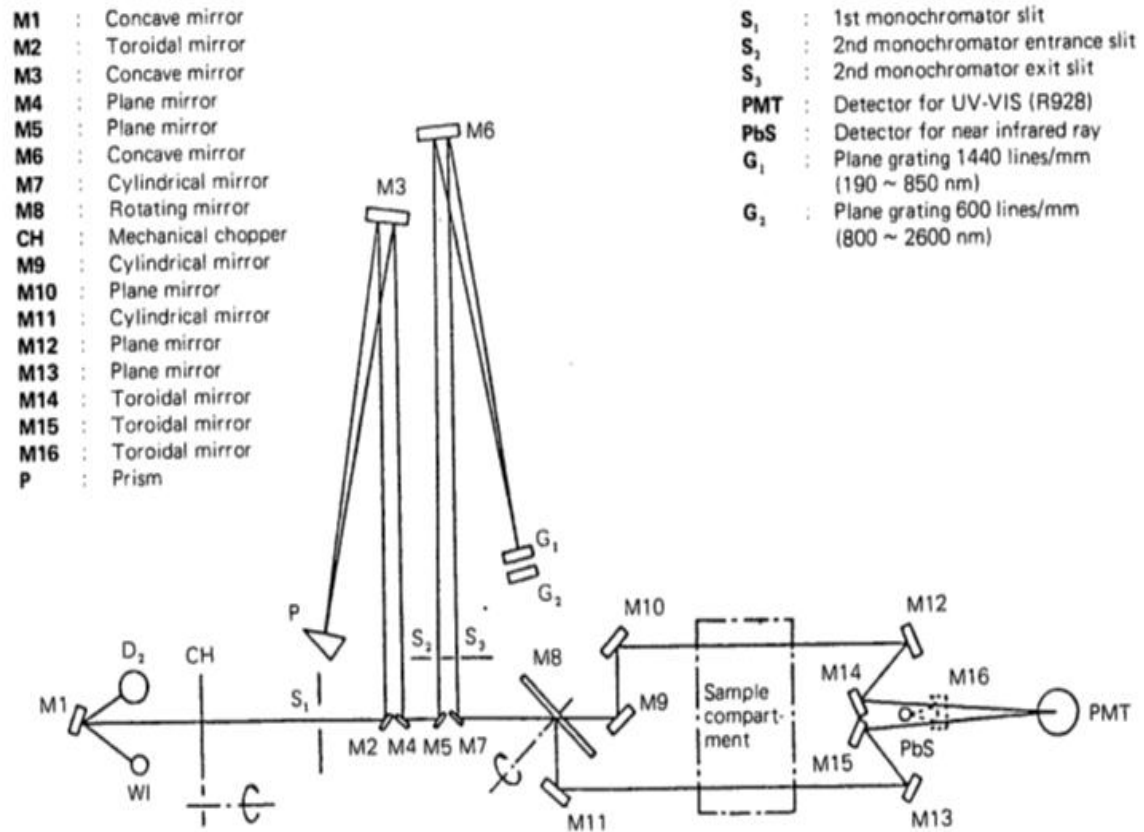


Figure 1. Optical diagram of the Hitachi U-3400 recording spectrophotometer

Since the illumination in the instrument is monochromatic, the spectral-power distribution of the source is not a factor as long as the intensity is sufficient to allow a satisfactory signal-to-noise ratio in the response [4]. Any attenuation of radiation other than that of the sample, such as lenses, mirrors, prisms, gratings and so on; are cancelled out because of the relativity of the measurement. The readout is displayed graphically with the percentage of transmittance plotted as a function of wavelength, and the absorbance (Abs) can be obtained by the log conversion of the transmittance.

$$A = \log_{10} \left(\frac{100}{T} \right) \dots \dots \dots (2)$$

Where *A* is the absorbance, in (Abs) and *T* is the transmittance in percentage (%).

4. REFERENCE MATERIAL

Laboratory spectrophotometers are available in a multitude of optical designs, specifications, spectral measurement ranges, capabilities, and operational configurations. Through calibration, equivalent measurements traceable to the national transmittance scale are possible regardless of the instrument design or region of the world where measurements are made. The practical way of calibrating the spectrophotometer will be to send it to the calibration laboratory. This process has proven to be costly and sometimes cumbersome. The availability of secondary reference standards is the ideal and most hassle-free approach to calibrating the instrument.

The wavelength and photometric reference standard of preference is the one which exhibits the most certified absorption peaks and transmittance/absorbance values in the same spectral region of the analytical method. For any wavelength standard, if no certified absorption peaks fall within the spectral region of interest, the use of a wavelength reference material that exhibits a certified absorption peak closest to the analytical wavelength of interest is acceptable [5]. Optical neutrality is a desirable spectral characteristic of a photometric reference material because it relaxes the restrictions on finite spectral bandwidth specification and wavelength accuracy to validate transmittance/absorbance measurements. A photometric reference material is considered optically neutral if its transmittance/absorbance spectrum varies little with wavelength.

Holmium oxide solution is superior to the holmium oxide glass filter as a UV/VIS wavelength reference material because of its higher certification accuracy and better UV spectral range coverage [5]. Having several certified absorption peaks in the 650 nm to 900 nm spectral range, the didymium oxide glass filter is preferred for validation of the wavelength scale in the near infrared spectral region. Solid photometric reference materials exhibit a higher degree of optical neutrality than photometric reference solutions [5]. The optical neutrality attribute is one of several reasons why solid photometric standards are preferred over photometric reference solutions for most photometric validations. Because of the lack of optical neutrality in their respective absorption spectra, the transmittance/absorbance certification for the photometric reference solutions is restricted to their fixed standard wavelengths.

5. VERIFICATION OF A REFERENCE TRANSFER SPECTROPHOTOMETER WITH PRIMARY STANDARD FILTERS

Verification of the wavelength accuracy and photometric scale on the Reference Transmittance Spectrophotometer is performed by using the standard reference material traceable to the National Physical Laboratory (NPL).

5.1. Baseline correction

It is important to perform a baseline correction before the start of every measurement, as this will affect the accuracy of the photometric scale of the spectrophotometer (**figure 2**) and it is always good practice to verify the wavelength scale of the instrument before the photometric scale and the linearity of the absorption scale can be performed.

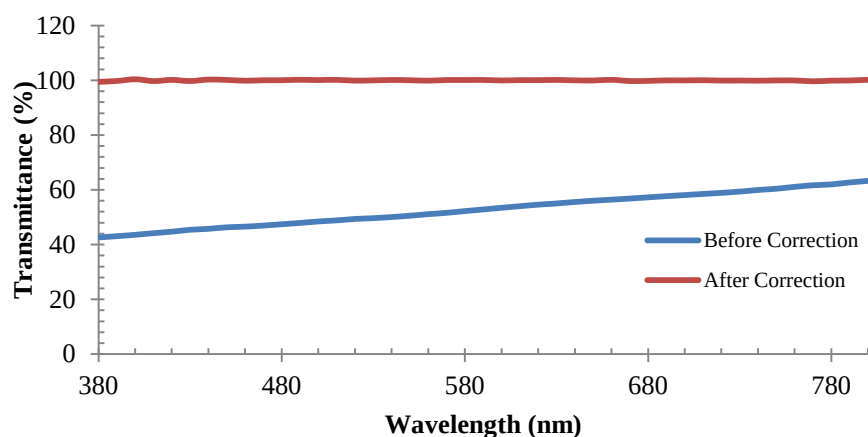
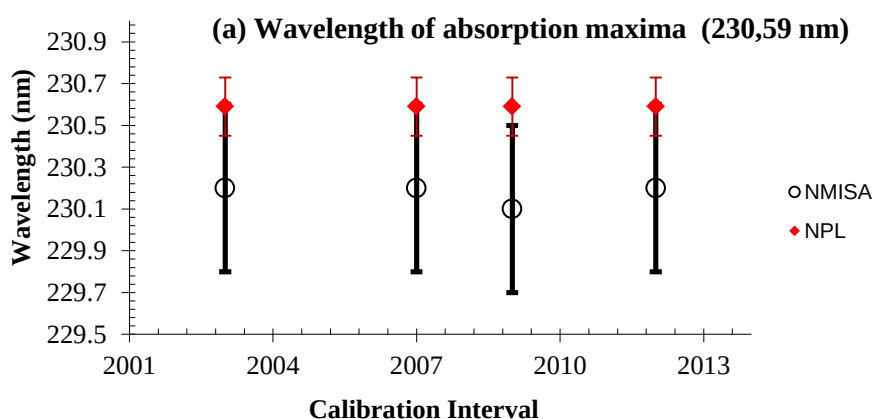
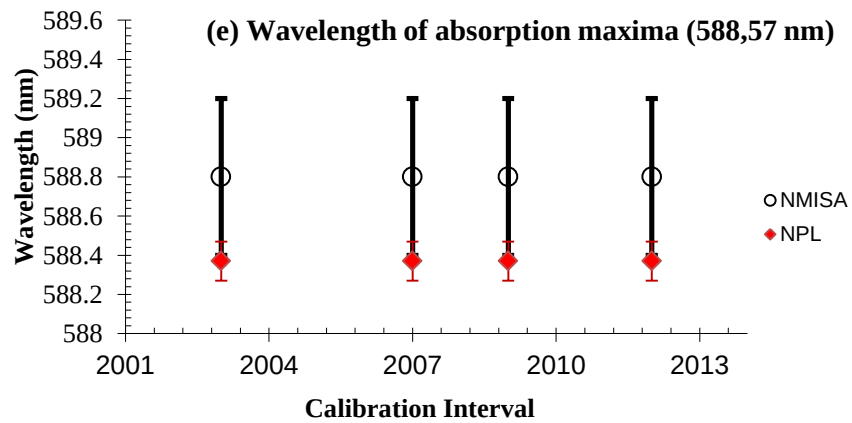
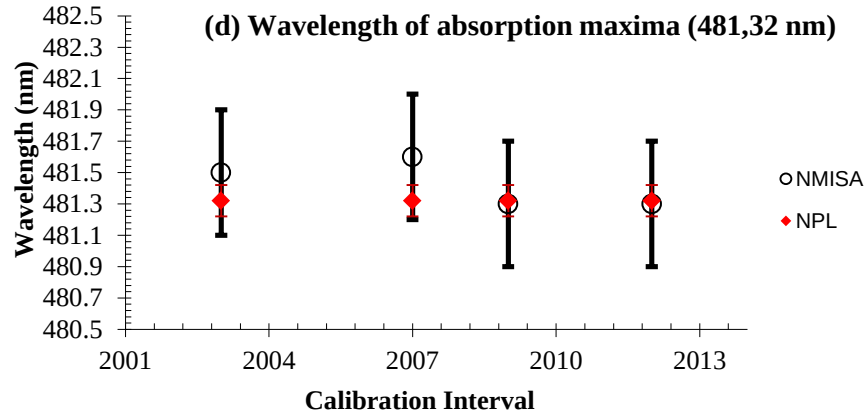
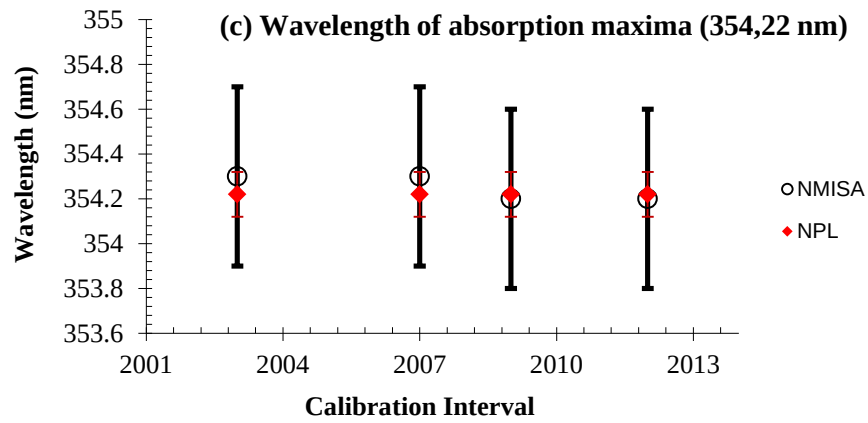
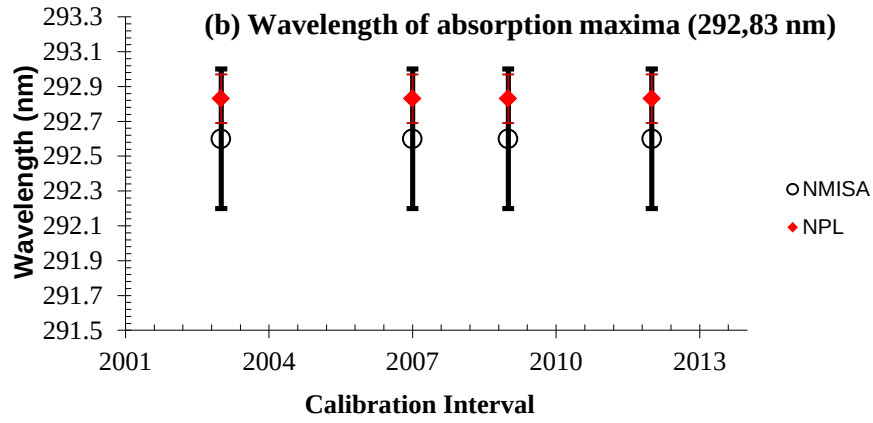


Figure 2. Effect of baseline correction

5.2. Primary Wavelength standard filters

To cover the wide spectral measurement range used in the different laboratory set-ups, the operational performance and data quality of the reference transmittance spectrophotometer is optimised to include a wide range of spectral regions. The results in **figure 3** show the nominal wavelength of absorption maximum of the primary wavelength standard filters covering the UV-Visible range. The measurements were performed at a bandwidth setting of 2 nm and the wavelength peak values obtained at each calibration interval were compared with the certified values obtained from the NPL calibration certificate. The red error bars indicate the expanded ($k=2$) claimed by the NPL, which was $\pm 0,14$ nm for wavelengths 230,59 nm and 292 nm. The black error bar indicate the expanded uncertainty ($k=2$) claimed by the NMISA which was 0.4 nm for all the indicated wavelengths.





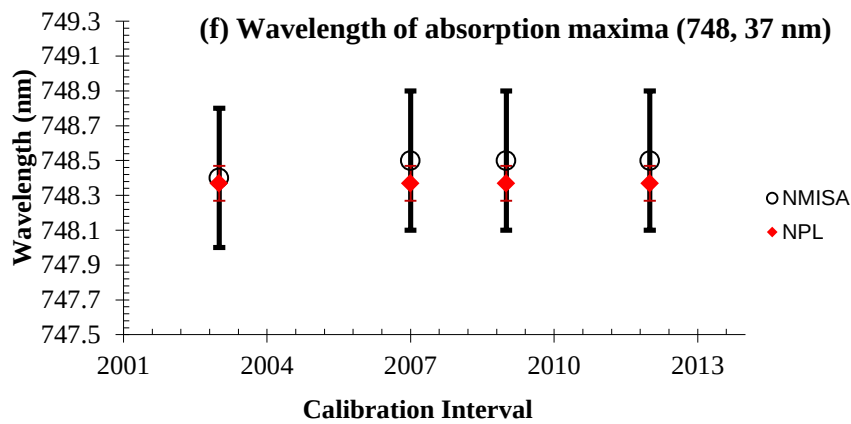


Figure 3. The nominal wavelength of selected absorption maxima of a holmium oxide filter (a-f) measured at the scheduled calibration interval.

The results show that the Reference Transmittance Spectrophotometer can measure the absorption peaks within the NPL claimed uncertainty values. The wavelength accuracy over scheduled intervals indicates that the performance of the moving mechanical components (which is responsible for positioning the monochromator) is stable.

5.3. Primary Neutral density filters

5.3.1. Precision and accuracy check

Like the wavelength standard filters, the photometric standard filters are available in different transmittance levels to cover a wide range of transmittance percentages. **Figure 4** shows the selected transmittance filters having different nominal transmittance levels (low, medium and high) used to verify the photometric scale on the reference spectrophotometer. The transmittance values presented are the average of the values measured over a period of nine years. The measurements were made at a spectral bandwidth setting of 2 nm and the values were compared with the one obtained from the calibration certificate. The uncertainty of the measurements claimed was $\pm 0.4\%$ transmittance.

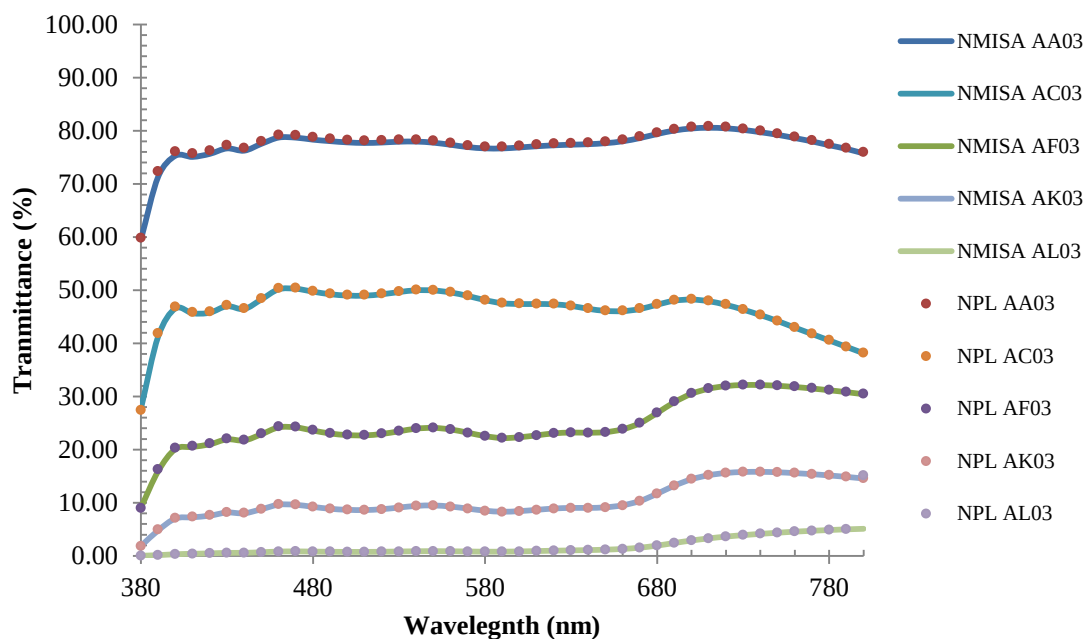


Figure 4. Average transmittance values of the selected filters measured at the scheduled calibration interval

As indicated in figure 4, the stability of the photometric scale over the scheduled calibration interval is adequate, thus the variations that might occur during the calibration of filters will not limit the precision and accuracy of the measurement data.

5.3.2. Linearity of the photo-detectors

Comprehensive validation of the spectrophotometer at periodic intervals includes a verification of photo-detector linearity using the full transmittance/ absorbance range of the photometric reference material standards. **Figure 5** show the neutral density filters with different transmittance/ absorbance levels used in the range of the calibration (near ultraviolet to visible). The measured transmittance values are plotted against the known NPL calibration values. The transmittance values vary from 0.08 %T to 80 %T and this corresponds to absorbance values ranges from 0.09 to 3 absorbance units. The results show that the photo-detector linearity is appropriate over the transmittance/absorbance range of the calibration.

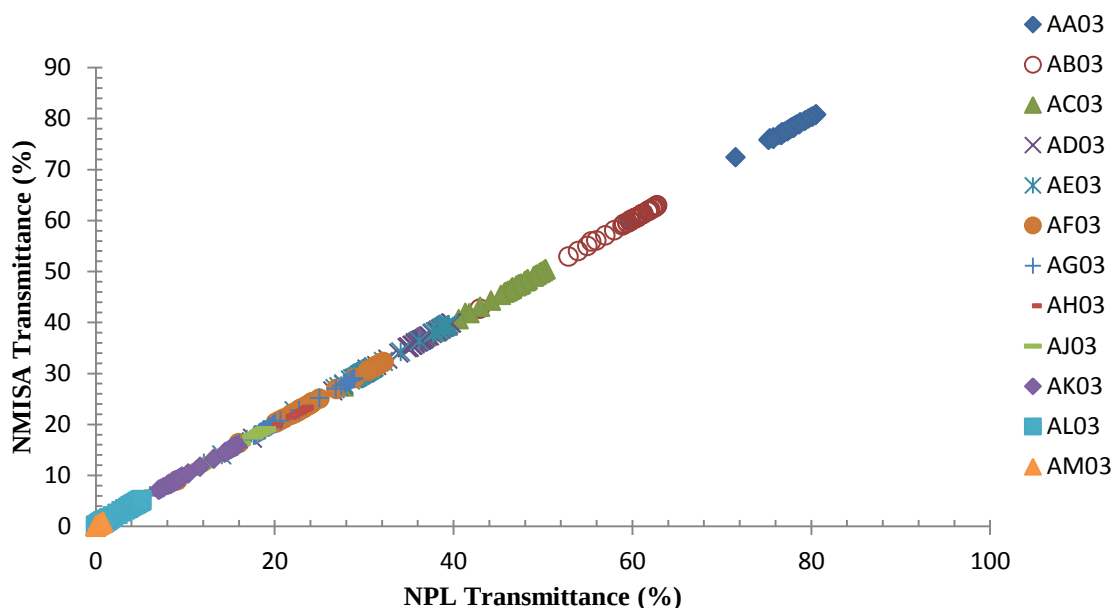


Figure 5. Linearity of the photo-detectors covering the near ultraviolet-visible region

6. DISSEMINATION OF NATIONAL TRANSMITTANCE SCALE

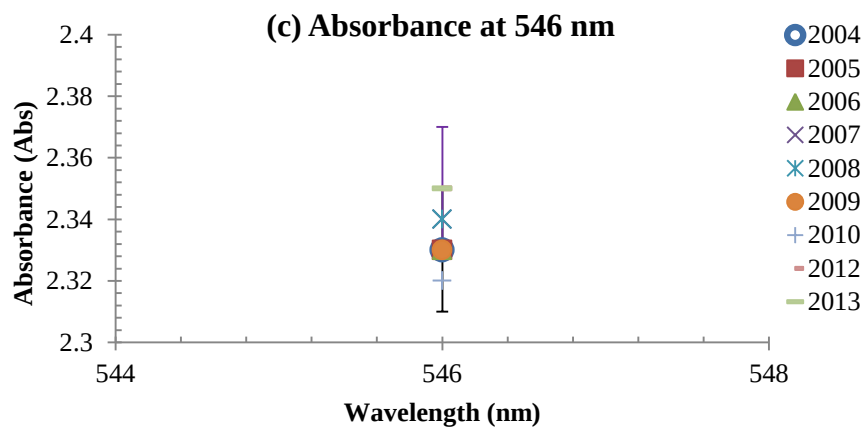
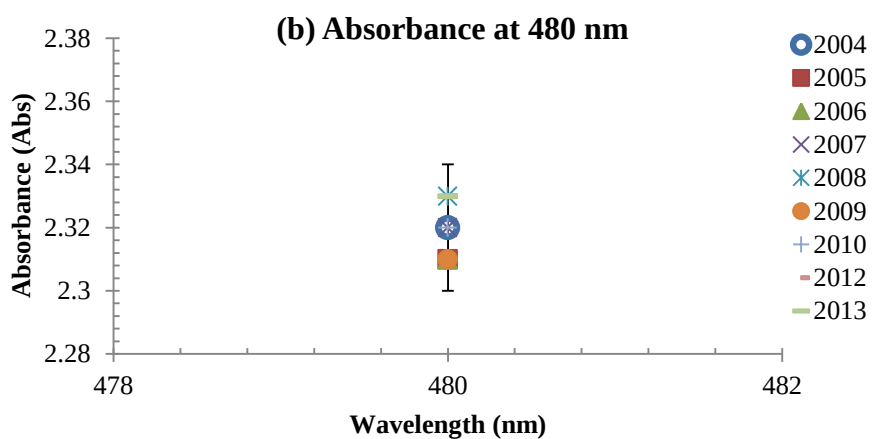
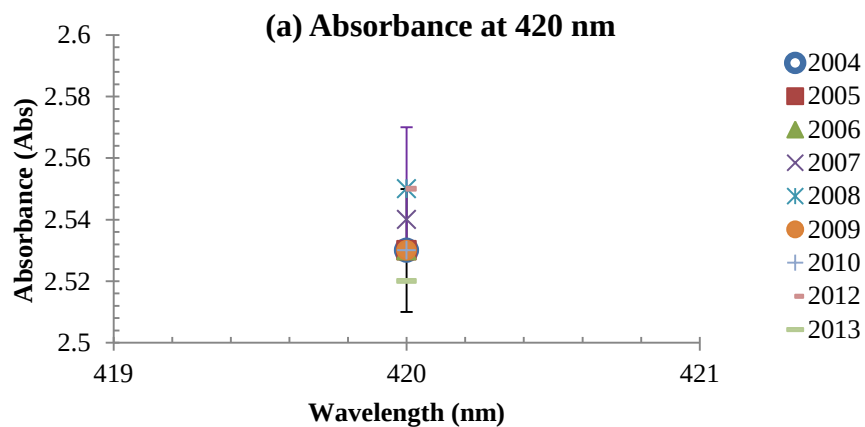
The periodic calibration of laboratories' spectrophotometers requires measurement of wavelength accuracy and absorbance. As mentioned in section 2, traceable measurements of the instrument are achieved by sending the secondary reference material standards for calibration. The calibrated filters are then used by the laboratory to verify the instrument used to perform the daily analysis.

6.1 Calibration of secondary neutral density filter

The absorbance of the neutral density filter was measured at wavelengths of 420 nm, 480 nm, 546 nm, 600 nm and 700 nm using a bandwidth of 2 nm. The result in **figure 6** shows the calibration of the filter over a period of nine years. The expanded uncertainties of the absorbance values measured are given in **Table 1**.

Table 1. Uncertainty of the measurement

Absorbance [Abs]	Expanded Uncertainty (k=2) [Abs]
$A < 1,0$	$\pm 0,005$
$1,0 \leq A < 2,0$	$\pm 0,01$
$2,0 < A < 2,5$	$\pm 0,02$



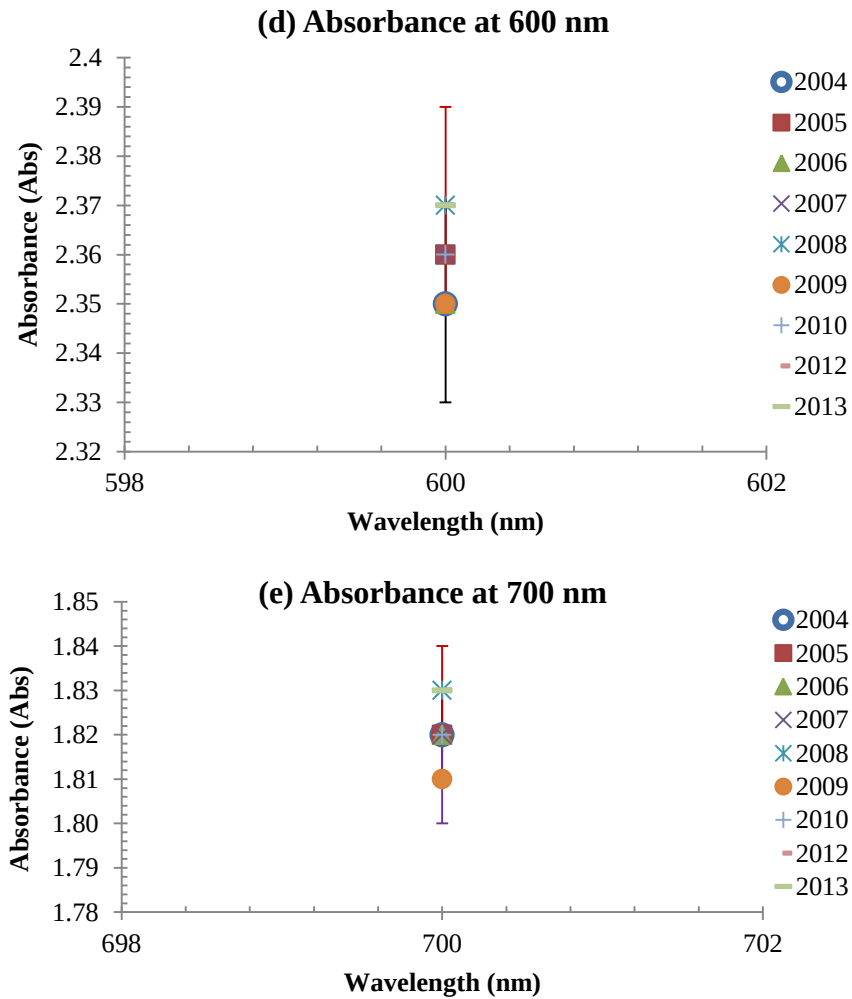


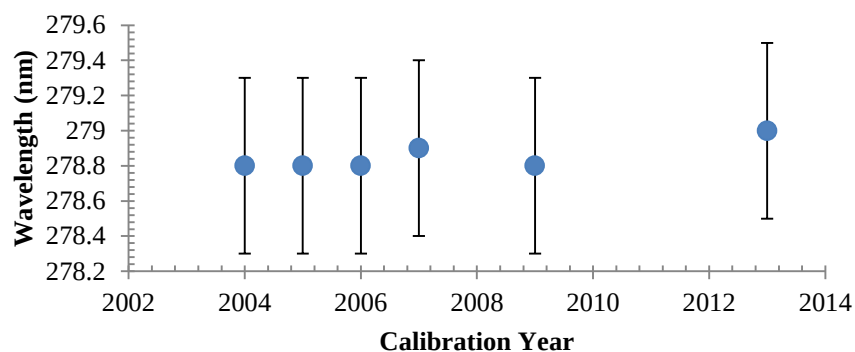
Figure 6. Absorbance values of the neutral density filter measured at the wavelength a) 420 nm, b) 480 nm, c) 546 nm, d) 600 nm, and e) 700 nm.

The result shows that the measured absorbance values done at the scheduled calibration intervals are within the claimed uncertainty shown in **Table.1**. Thus the spectrophotometer is capable of reproducing the measured values through the years.

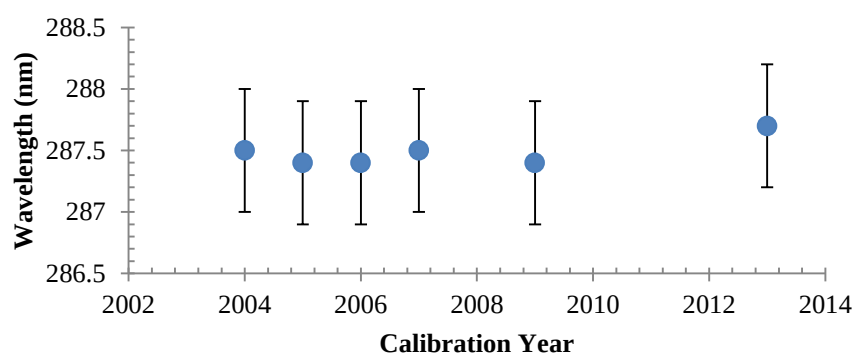
6.2 Calibration of secondary Holmium Oxide filter

The spectral absorbance of the Holmium Oxide filter was measured from 200 nm to 800 nm using a bandwidth of 2 nm. The result in **figure 7** shows nominal wavelength of selected absorption maxima. The expanded uncertainty ($k=2$) of the wavelength values was $\pm 0,4$ nm.

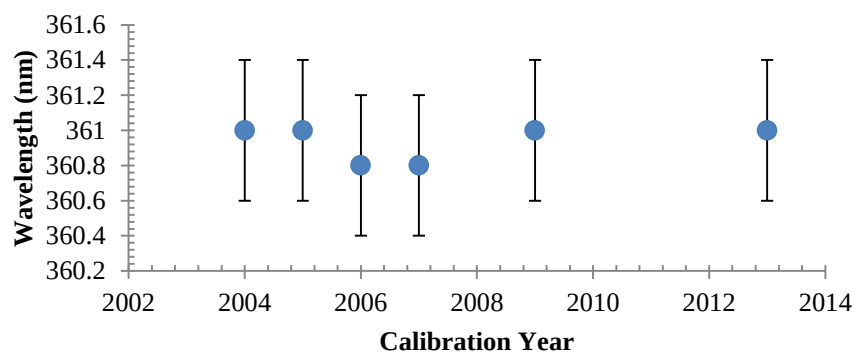
(a) Wavelength of absorption maxima (278,8 nm)



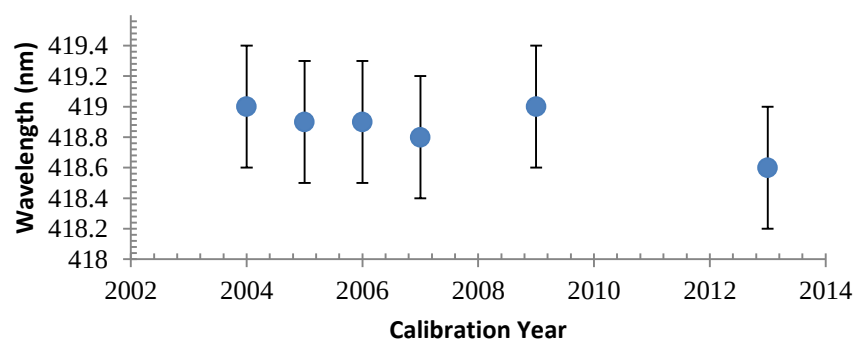
(b) Wavelength of absorbtion maxima (287,5 nm)



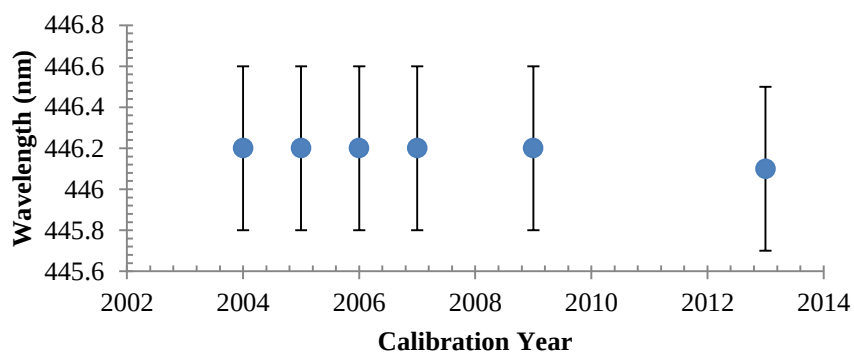
(c) Wavelength of absorbtion maxima (361,0 nm)



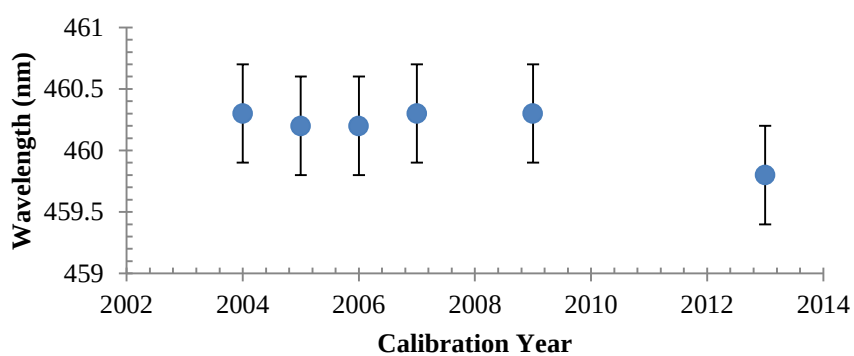
(d) Wavelength of absorbtion maxima (419,0 nm)



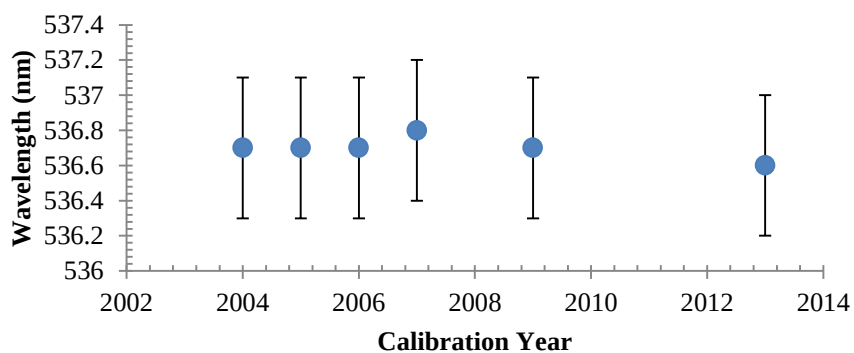
(e) Wavelength absorption maxima (446,2 nm)



(f) Wavelength absorption maxima (460,3 nm)



(g) Wavelength absorption maxima (536,7 nm)



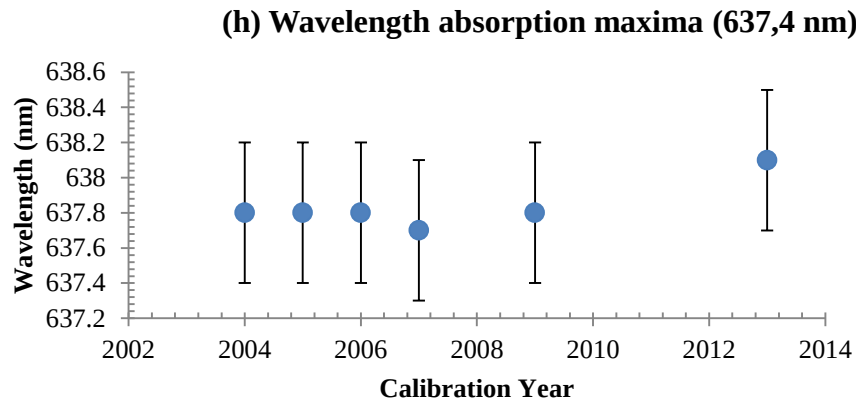


Figure 7. The nominal wavelength of selected absorption maxima of a holmium oxide filter (a-h) measured at the scheduled calibration interval.

The result shows that the measured peak wavelength values are within the claimed uncertainty. Thus the spectrophotometer is capable of reproducing the measured values through the years.

7. CONCLUSION

The verification tests in **Section 5.2 and 5.3** facilitate the acceptance and use of the reference transmittance spectrophotometer to disseminate the transmittance scale with great confidence. In a laboratory setting, the performance verification (PV) tests on the field spectrophotometer will be performed using secondary reference materials (RMs) and the results in section 6.1 and 6.2 indicates that they are traceable to primary reference standards.

It is important that reference materials used are accurate and certified in the spectral wavelength region of analytical interest. This is important because the operational performance and data quality of any given spectrophotometer may be optimal in certain spectral regions, but compromised in other spectral regions. The use of traceable secondary reference materials has also been considered an important tool to investigate potential instrument-related systematic errors that may impact on the quality and reliability of the data. Partnering with a reputable UV/VIS measurement science and standards organisation will reinforce or enhance the laboratory's quality consciousness and reputation, which will further support its commitment to analytical excellence in the competitive global marketplace.

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